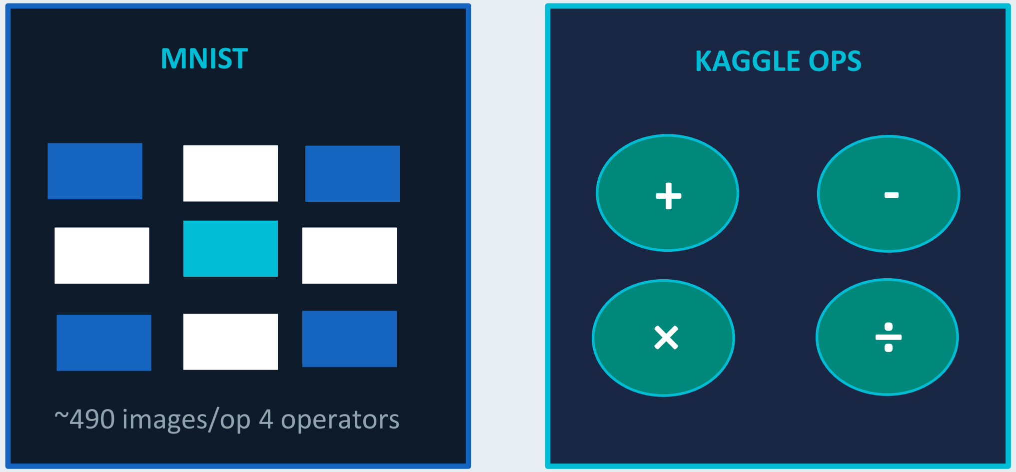
Architecture · Training · Evaluation

14	**177K**	**64192**
Classes	Params	Training Samples

Dataset & Problem Setup

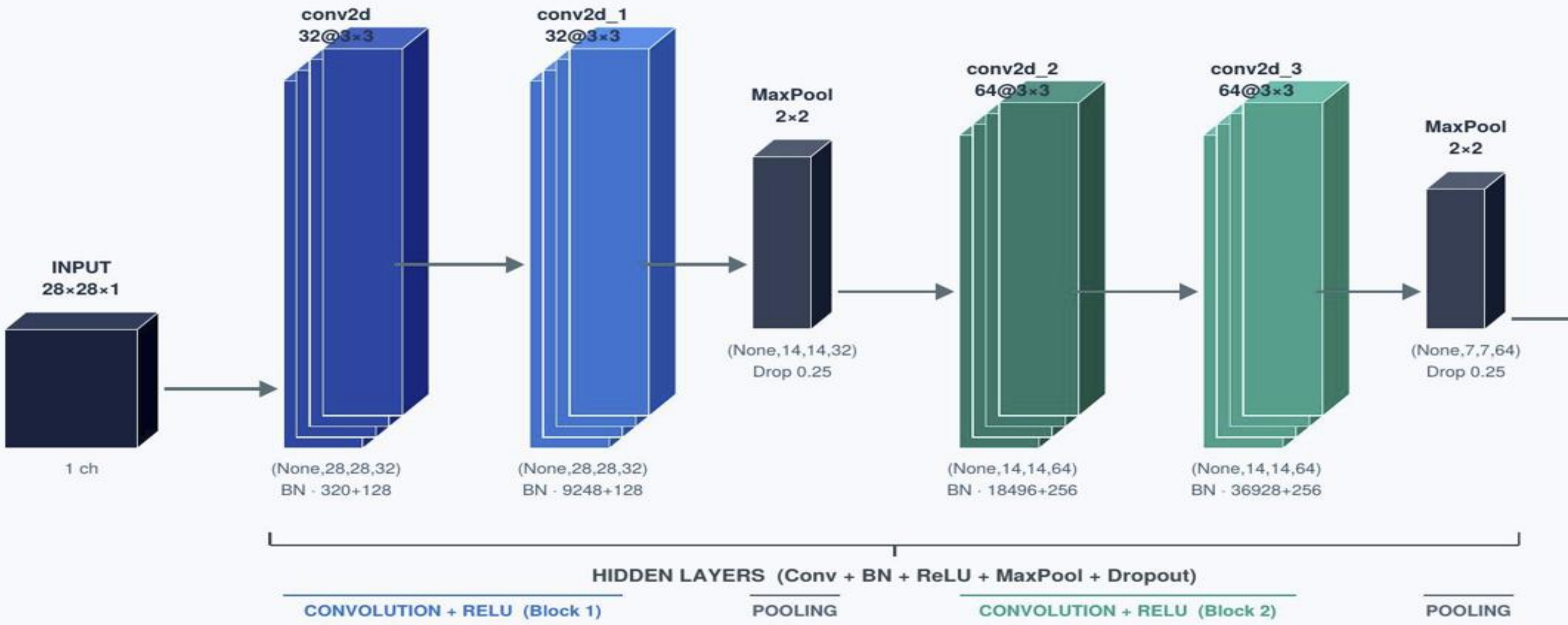
DATA SOURCES



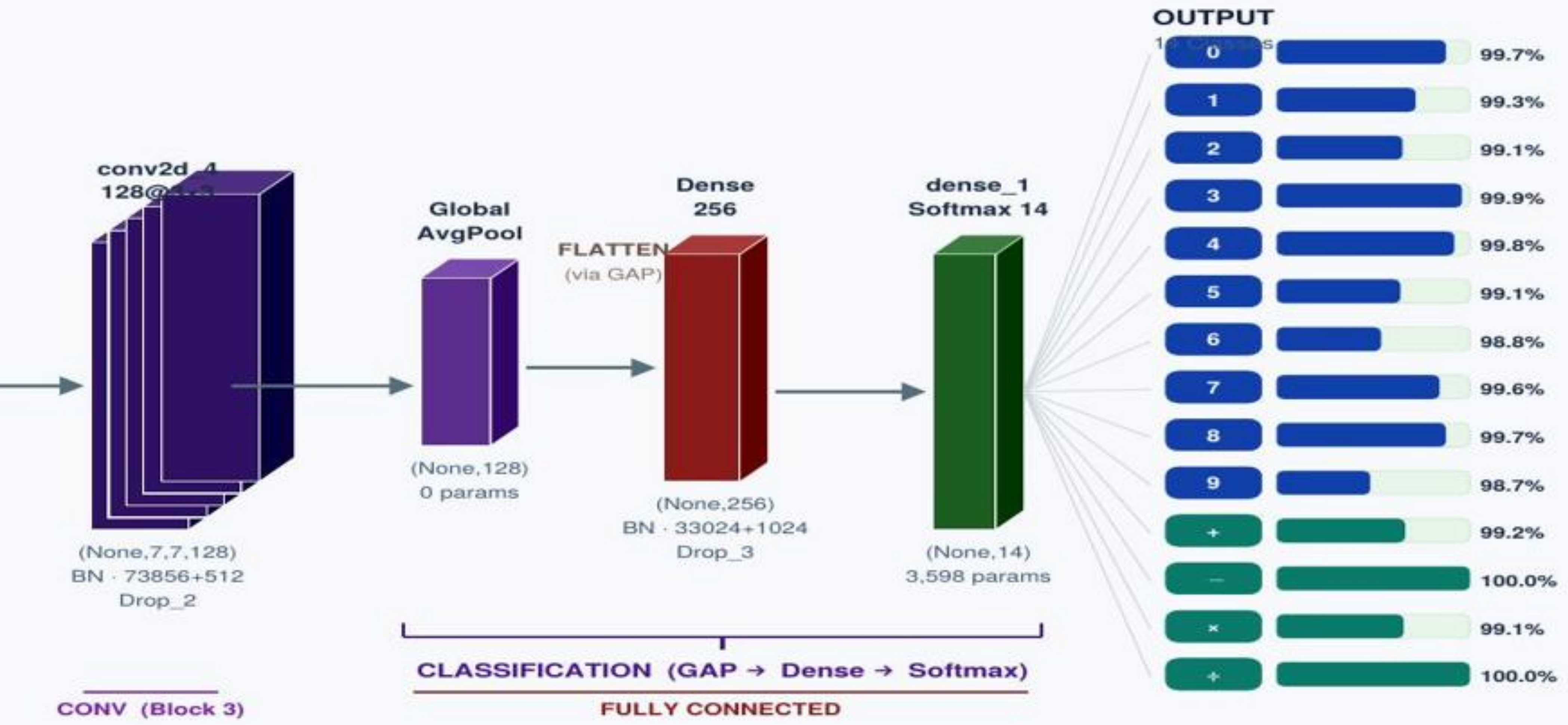
Total: 14 Classes | Balanced via augmentation

- 28×28: Grayscale input
- Normalize: Pixel ÷ 255
- Elastic: Deform augment
- Balance: Target per class

Convolution Neural Network (CNN) Architecture



Convolution Neural Network



Key design choices

Batch Normalization Rescales layer outputs to mean zero and unit variance. Accelerates training by up to 3× and provides mild regularization.

MaxPooling (2×2) Downsamples spatial dimensions to reduce computation costs while making detected features invariant to small translational shifts.

Dropout Regularization Randomly silences 25–50% of neurons during training. This prevents memorization and forces robust, distributed learning.

GlobalAveragePooling Replaces the Flatten layer by averaging feature maps into a compact vector, dramatically reducing model parameter count.

Training Configuration & Data Augmentation

HYPERPARAMETERS

Optimizer	AdamW lr=1e-3 wd=1e-4
Loss	Categorical Cross-Entropy
Batch Size	128
Epochs	20 (early stop @ 18)
Best Epoch	Epoch 13
Callbacks	EarlyStopping + ReduceLROnPlateau

LR SCHEDULE

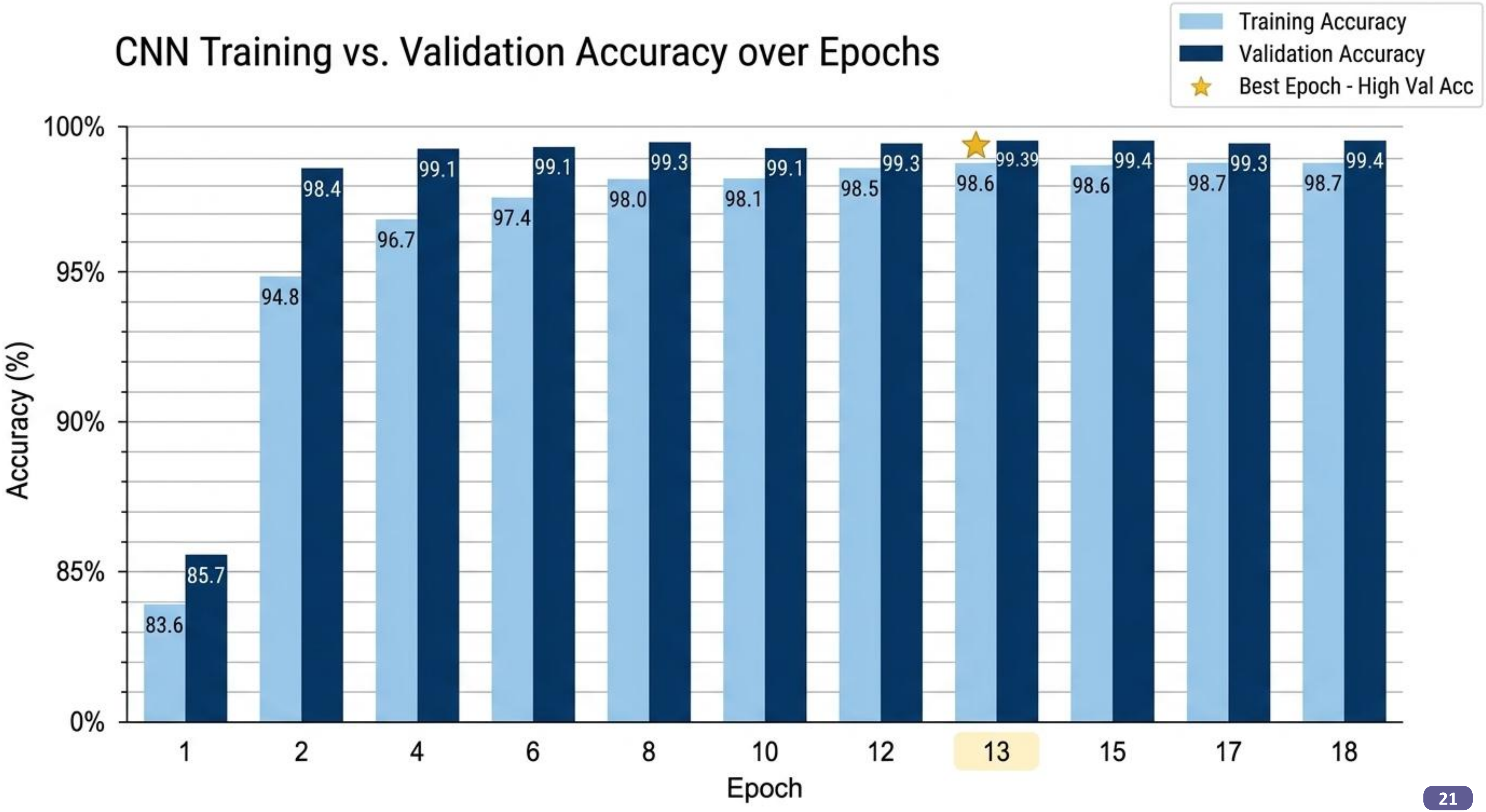
1e-3	5e-4 (ep6)	2.5e-4 (ep10)	1.25e-4 (ep12)	6.25e-5 (ep17)

AUGMENTATION PIPELINE

Random Rotation	±8% (28.8°)	±8% (28.8°)
Random Translation	±8% H & V	±8% H & V
Random Zoom	±10%	±10%
Elastic Transform	α=8 σ=4 p=0.7	α=8 σ=4 p=0.7
Pixel Normalize	÷ 255 → [0,1]	÷ 255 → [0,1]

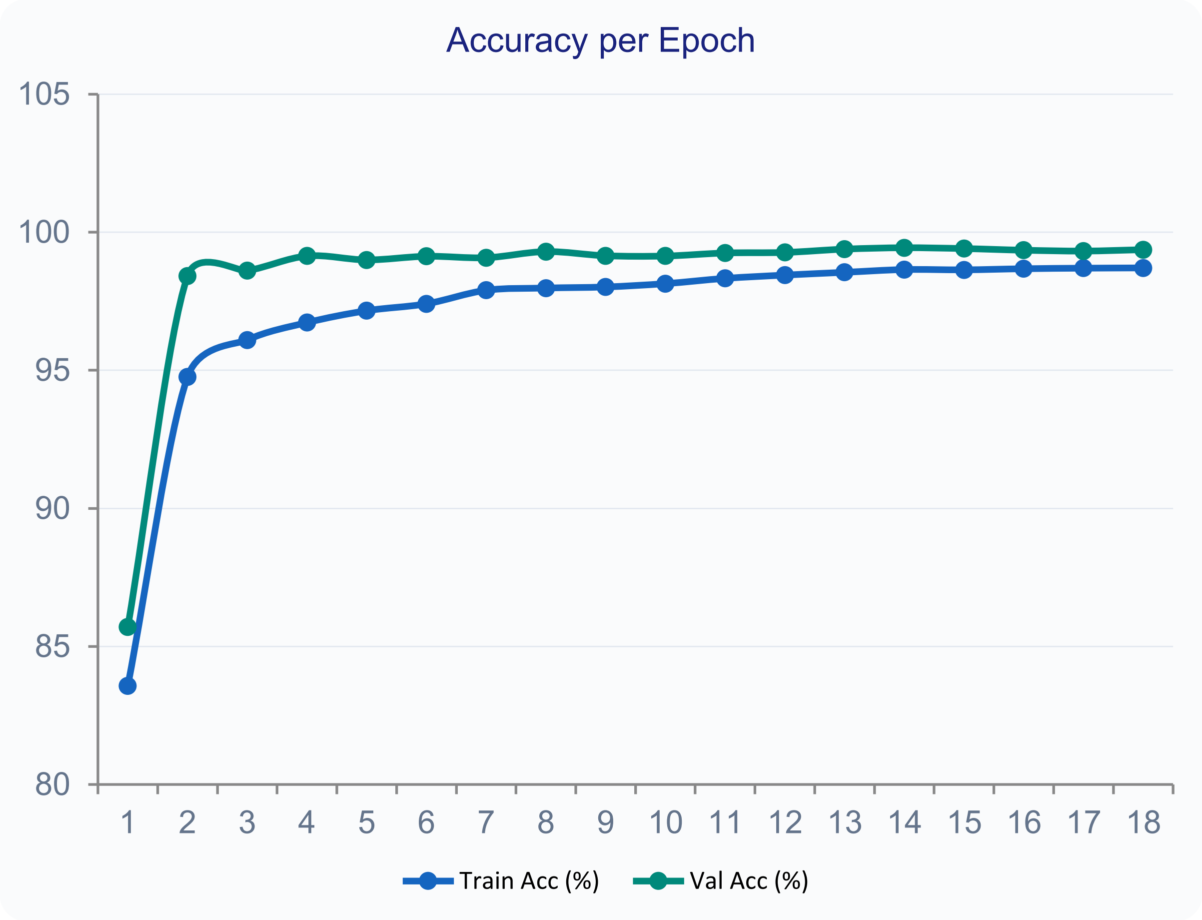
Training Progress

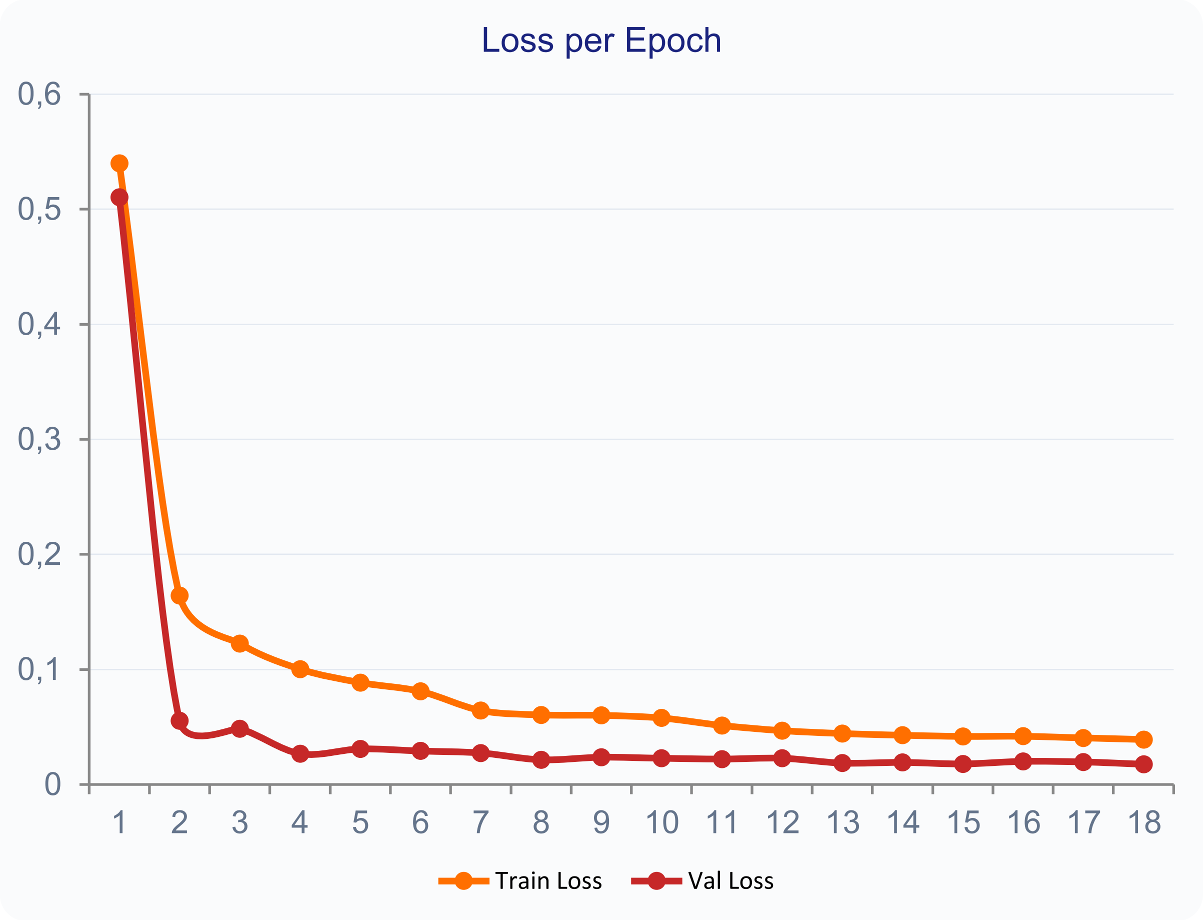
Epoch	Training Acc	Validation Acc	Learning Rate	Notable event
1	83.6%	85.7%	0.001000	Initial run — model still learning basics
2	94.8%	98.4%	0.001000	Huge jump — core patterns learned quickly
4	96.7%	99.1%	0.001000	Steady improvement
6	97.4%	99.1%	0.000500	LR halved for the first time
8	98.0%	99.3%	0.000500	Finer adjustments with lower LR
10	98.1%	99.1%	0.000250	LR halved again
12	98.5%	99.3%	0.000125	LR halved again — very fine-tuning
13	**98.6%**	**99.39%**	**0.000125**	**★ Best epoch — highest validation accuracy**
15	98.6%	99.4%	0.000125	Slight further improvement in val acc
17	98.7%	99.3%	0.0000625	LR halved for the fourth time
18	98.7%	99.4%	0.0000625	EarlyStopping fires — training ends here



CNN Training vs. Validation Accuracy over Epochs
Training Accuracy
Validation Accuracy
Best Epoch - High Val Acc
100%
95%
90%
85%
0%
Accuracy (%)
83.6
85.7
94.8
98.4
96.7
99.1
97.4
99.1
98.0
99.3
98.1
99.1
98.5
99.3
98.6
99.39
98.6
99.4
98.7
99.3
98.7
99.4
1
2
4
6
8
10
12
13
15
17
18
Epoch

Training Progress : Accuracy & Loss Curves





Fast convergence: val acc jumped from 85.7% → 98.4% in just 2 epochs

Best epoch: 13 (val acc 99.39%, val loss 0.0186)

Early stopping triggered at epoch 18, no overfitting observed

ReduceLROnPlateau halved LR at epochs 6 / 10 / 12 / 17

Confusion Matrix

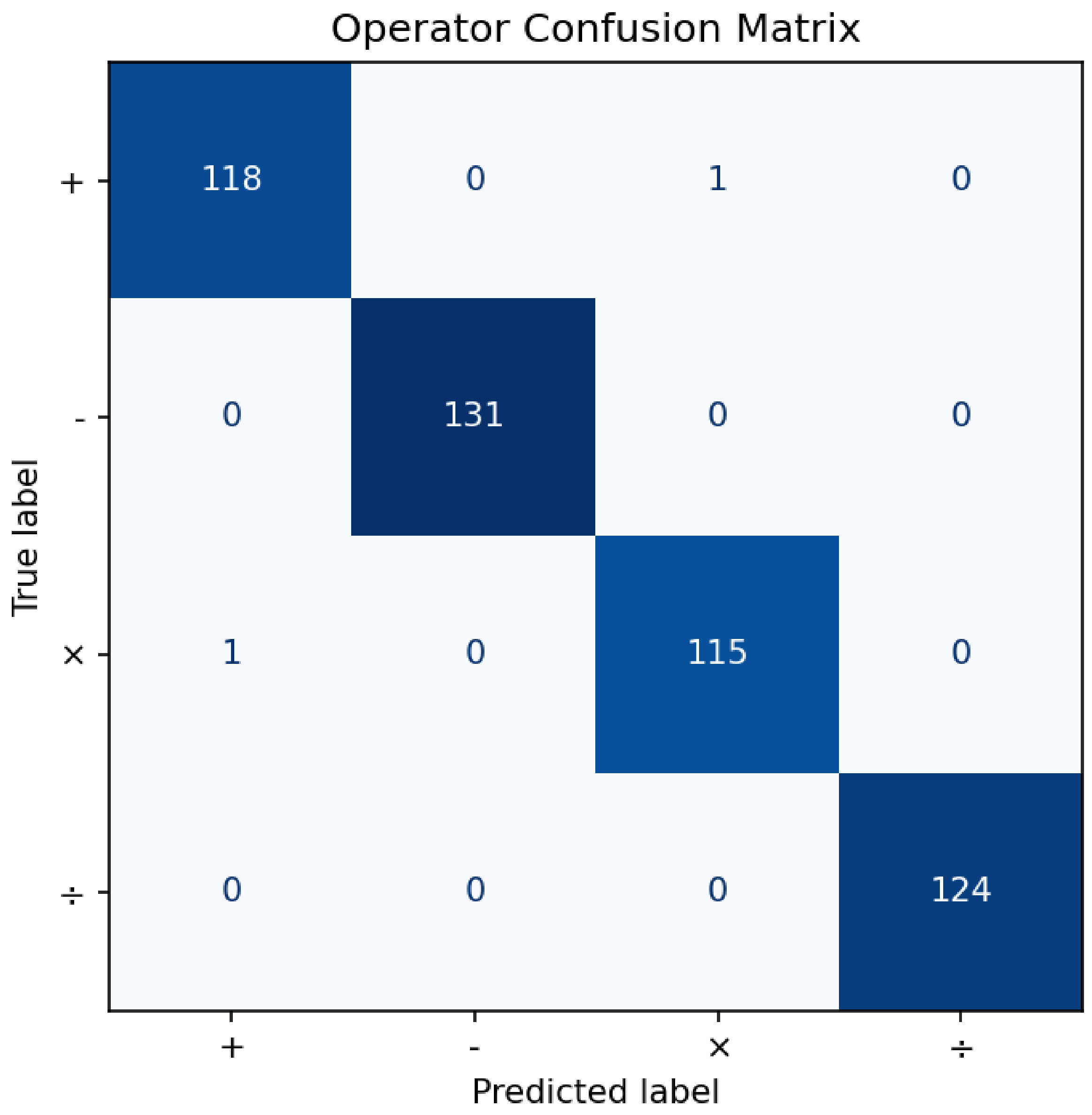
Confusion Matrix (Counts)

True label \ Predicted label	0	1	2	3	4	5	6	7	8	9	+	-	×	÷
0	977	0	1	1	0	1	0	0	0	0	0	0	0	0
1	0	1127	0	0	1	0	0	6	0	0	0	0	0	1
2	1	0	1023	1	0	0	0	6	1	0	0	0	0	0
3	0	0	0	1009	0	1	0	0	0	0	0	0	0	0
4	0	0	0	0	980	0	0	0	0	2	0	0	0	0
5	0	0	0	4	0	884	3	1	0	0	0	0	0	0
6	3	3	0	0	1	2	947	0	2	0	0	0	0	0
7	0	1	3	0	0	0	0	1024	0	0	0	0	0	0
8	0	0	0	0	1	0	0	1	971	0	0	0	0	1
9	0	0	0	1	7	1	0	3	1	996	0	0	0	0
+	0	0	0	0	0	0	0	0	0	0	118	0	1	0
-	0	0	0	0	0	0	0	0	0	0	0	131	0	0
×	0	0	0	0	0	0	0	0	0	0	1	0	115	0
÷	0	0	0	0	0	0	0	0	0	0	0	0	0	124

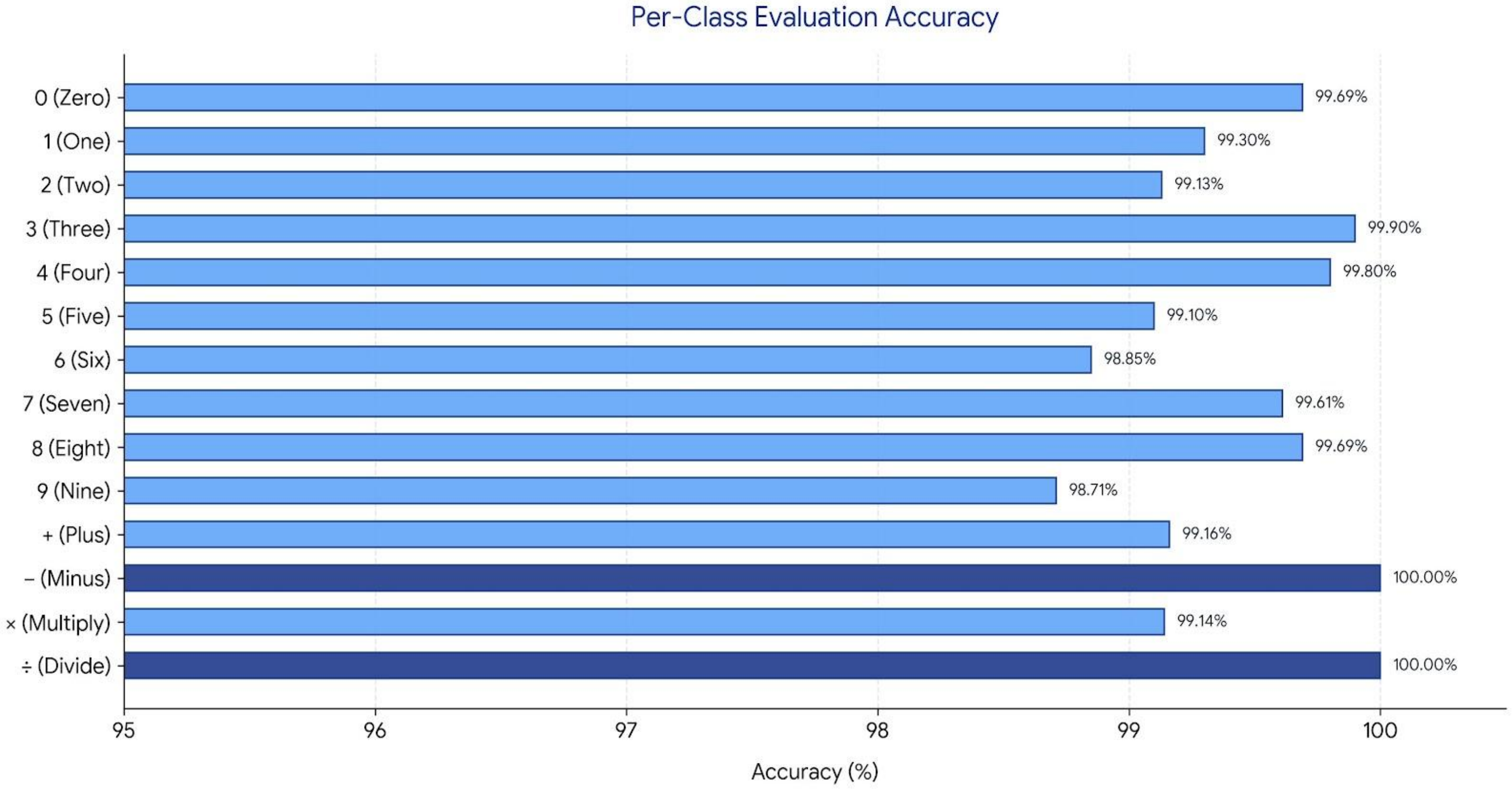
Confusion Matrix (Normalized)

True label \ Predicted label	0	1	2	3	4	5	6	7	8	9	+	-	×	÷
0	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1	0.00	0.99	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00
2	0.00	0.00	0.99	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00
3	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
4	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
5	0.00	0.00	0.00	0.00	0.00	0.99	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
6	0.00	0.00	0.00	0.00	0.00	0.00	0.99	0.00	0.00	0.00	0.00	0.00	0.00	0.00
7	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00
8	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00
9	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.99	0.00	0.00	0.00	0.00
+	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.99	0.00	0.01	0.00
-	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00	0.00
×	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.99	0.00
÷	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00

Operator Confusion Matrix



Test Accuracy



Prediction Pipeline & Safe Evaluation

Draw
User draws on
Streamlit canvas

Preprocess
Resize 28×28
Normalize ÷ 255

CNN Infer
Forward pass
best_model.keras

Softmax
14-class probs
Argmax → label

Evaluate
Parse equation
Compute result

SAFE EVALUATION DESIGN

- ✓ Compile guard: model.compile() enforced before evaluate() call — catches ValueError before production
- ✓ Class-split eval: digit accuracy (99.38%) and operator accuracy (99.59%) measured independently to catch domain imbalance
- ✓ Round-trip check: best_model.keras reloaded and re-evaluated after save to confirm lossless serialization
- ✓ Division guard: eval logic checks divisor ≠ 0 before computing; invalid expressions return 'Error' not exception

SUMMARY

Architecture

5 Conv layers (32→32→64→64→128)

GlobalAvgPool → Dense 256 → Softmax 14

177K params, only 694 KB

Data Strategy

MNIST (60K) + Kaggle operators (~490/class)

Auto-balanced · Elastic deform · Online aug

Heavy augmentation reduces overfitting

Training

AdamW + ReduceLROnPlateau (×0.5 on plateau)

EarlyStop @ ep18, best weights restored ep13

Fast convergence: 98.4% val acc by epoch 2

Results

Overall: 99.39% | Digits: 99.38%

Operators: 99.59% | − & ÷: 100%

Consistent across all 14 classes ≥ 98.7%